Review notes · for you

# Data science and society: unit check

Data science · Level 4 · Unit check

| Time | Materials |
| --- | --- |
| 25 min | Paper and a pen. No device needed. |

## Learning goals

- Samples and margins
- Finding groups
- Fair models
- Deciding with data

## Activity: Revise in pairs

1. Pairs (2 min): Work with a partner and share one sheet for your answers.
2. Quiz (15 min): Read each question, agree on an answer and say why.
3. Check (8 min): Compare with the answers and redo the ones you got wrong.

## Discussion

- Which idea from this unit will you use first?
- Which app or survey will you check for fairness?
- Which hunch will you test with data?

## Check yourself

- One small fair trial I could run at my school or college, and its comparison group.

**Try it this week**

Find one way data is used in your town or school, and ask: who is missing, is it fair, and who decides?

Quiz sheet

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Name _____ Class _____ Date _____

**1** **s = 10, n = 25.**

The standard error is _____ .

**Word bank:** 0.4 · 2

**2** **A larger sample fixes selection bias.**

◯ True ◯ False

**3** **To halve the standard error, n must be multiplied by…**

◯ 8 ◯ 2 ◯ 4

**4** **Order the k-means loop.**

☐ Update centroids

☐ Initialise centroids

☐ Repeat to convergence

☐ Assign points

**5** **Different starting centroids can give different k-means results.**

◯ True ◯ False

**6** **False negatives by group.**

Group A misses 2 of 20 positives ( _____ %); group B misses 8 of 20 ( _____ %).

**Word bank:** 2 · 10 · 40 · 8

**7** **Which can carry bias into a model? Choose all that apply.**

☐ Using a test set ☐ Historically biased labels ☐ Under-represented groups ☐ Proxy features

**8** **Match each term to its meaning.**

| | | |
|---|---|---|
| Confounder | • • • • | Random assignment |
| RCT | • • • • | Treated change minus control change |
| Difference-in-differences | • • • • | Affects treatment and outcome |

**9** **Treated: 20 → 14. Control: 19 → 17.**

The difference-in-differences effect is _____ .

**Word bank:** -6 · -4

**10** **Who should make the final decision?**

◯ Whoever collected the data ◯ Accountable people, informed by data and stakeholders ◯ The model

Check your answers

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1. **The standard error is 2.**
   10 ÷ √25 = 10 ÷ 5 = 2.

2. **False**
   Size reduces random error, not bias.

3. **4**
   SE scales with 1 ÷ √n.

4. **Top to bottom: 3, 1, 4, 2**
   Assign and update alternate.

5. **True**
   It finds local optima.

6. **Group A misses 2 of 20 positives (10%); group B misses 8 of 20 (40%).**
   2 ÷ 20 = 10%, 8 ÷ 20 = 40%.

7. **Proxy features; Under-represented groups; Historically biased labels**
   A test set helps detect bias; it doesn't cause it.

8. **Confounder — Affects treatment and outcome**
   **RCT — Random assignment**
   **Difference-in-differences — Treated change minus control change**
   Tools for causal claims.

9. **The difference-in-differences effect is -4.**
   (14 − 20) − (17 − 19) = −6 − (−2) = −4.

10. **Accountable people, informed by data and stakeholders**
    Data informs; people decide and are accountable.